Aura: An AI-based Carbon Footprint Calculator and Short-Term Predictor for India

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***Abstract*—This paper presents Aura — an end-to-end platform for calculating and predicting personal carbon footprints tailored to the Indian context. Aura integrates a web frontend, a Prisma-backed emission-factor database seeded with India-relevant factors, a deterministic calculator for user activity inputs, and a lightweight, explainable time-series predictor. The design prioritizes reproducibility, transparency, and accessibility for non-expert users. The system enables individuals to understand and monitor their environmental impact, empowering more sustainable behavior and aligning with India's climate goals.**

***Index Terms*—carbon footprint; emission factors; India; sustainability; prediction; Prisma; Next.js; time-series**

I. Introduction

Greenhouse gas emissions from daily activities contribute significantly to global climate change. In India, with rapid urbanization, increasing energy demand, and lifestyle diversification, understanding personal and regional carbon emissions has become critical. While several global carbon calculators exist, they often rely on datasets and emission factors not representative of India's energy mix, transportation systems, and cultural habits. This leads to inaccurate estimations that limit their relevance for Indian users.

Aura addresses this gap by developing a localized and transparent platform that combines authoritative Indian data sources with modern web and data science frameworks. The tool allows users to estimate and predict their short-term carbon footprint through intuitive interfaces and explainable models. By contextualizing sustainability data for Indian households and individuals, Aura supports both behavioral change and broader awareness campaigns aligned with the Government of India's commitments to the Paris Agreement and the National Action Plan on Climate Change (NAPCC).

II. Related Work

Previous studies have explored static and dynamic carbon footprint estimation methods. However, many tools lack regional customization and transparency in emission factor derivation. Aura leverages datasets from IPCC, the Central Electricity Authority (CEA), and the Ministry of Environment, Forest and Climate Change (MoEFCC), ensuring scientific accuracy and contextual relevance. Reproducibility and provenance are central design principles, overcoming the opacity seen in several commercial calculators.

III. System Architecture and Implementation

Aura follows a modular and scalable architecture. The frontend, built with Next.js and Tailwind CSS, provides a responsive user experience optimized for both web and mobile platforms. The backend uses Node.js with Prisma ORM for robust database management. A structured emission-factor database stores values categorized by sector, region, and activity type. A forecasting microservice runs lightweight predictive models that estimate near-future emissions based on user history and population averages when individual data is limited.

IV. Deterministic Emission Calculator

The deterministic calculator serves as the backbone of Aura. It standardizes user inputs (such as kilometers driven, electricity consumed, or meals eaten), maps them to relevant emission factors, and computes emissions in CO_2e. Each calculation is traceable, allowing reproducibility and transparency. The modular design supports the addition of new categories or regional emission updates without breaking the workflow.

V. Prediction Model

Aura integrates a linear regression-based predictor for short-term emission forecasting. The model uses historical patterns to estimate next-month emissions, incorporating confidence intervals to convey uncertainty. For users with sparse data, it defaults to population-weighted averages. This approach ensures reliability while remaining explainable, in contrast to opaque black-box neural models.

VI. Data and Emission Factors

Emission factors are derived from authoritative Indian sources covering transportation fuels, grid electricity, LPG and

PNG usage, and common food items. Each factor includes source metadata and version control for reproducibility. The dataset is periodically updated to reflect changes in India's national emission database and energy mix.

VII. Experiments and Evaluation

Aura was evaluated using synthetic and anonymized datasets reflecting diverse Indian consumption patterns. The prediction component was tested with standard metrics such as RMSE, MAE, and MAPE. The deterministic calculator underwent unit validation to ensure consistency and correctness across all categories. User testing demonstrated that localized factors significantly improved perceived accuracy and engagement compared to international calculators.

VIII. Results

Results show that Aura's forecasting module achieves low RMSE and MAE scores for short-term trends, outperforming naive and static baselines. The transparent nature of the regression model helps users understand emission drivers and seasonal variations. Qualitative user feedback highlighted the platform's usability and contextual clarity.

IX. Discussion

Aura demonstrates how contextualized, transparent, and reproducible carbon accounting tools can foster environmental literacy in rapidly developing nations. In India, where sustainability awareness is often hindered by lack of accessible data, Aura bridges the gap between scientific research and public engagement. The tool not only quantifies emissions but also facilitates behavioral insights, enabling individuals to align their choices with national sustainability goals. This aligns with broader efforts such as India's "LiFE" (Lifestyle for Environment) movement, which emphasizes the importance of sustainable personal choices.

X. Future Work

Future work includes integrating satellite-based emission data for regional granularity, incorporating life-cycle assessment (LCA) factors for consumer goods, and developing a federated privacy-preserving learning framework to improve model generalization while safeguarding user data. The planned LSTM-based modules will capture seasonal and behavioral trends for long-term prediction.

XI. Limitations and Ethical Considerations

Limitations include potential inaccuracies in user-reported data, reliance on static emission factors, and challenges in modeling nonlinear behavior. Ethically, Aura is designed with privacy as a first-class principle — it avoids storing sensitive personal information and ensures user data anonymization. Transparency in model outputs and uncertainty reporting is maintained to avoid misleading interpretations. The platform supports open data collaboration, aligning with responsible AI and climate research ethics.

XII. Conclusion

Aura represents a step toward contextual, transparent, and explainable carbon footprint assessment in India. By focusing on reproducibility and accessibility, it empowers citizens to make informed sustainability decisions while contributing to national and global climate targets. Aura illustrates that localized digital tools can amplify environmental awareness and bridge the gap between individual responsibility and collective climate action.

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